PROJECT ESE

DEMENTIA DETECTION

Using EEG Signal Analysis & Deep Learning

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PROJECT RECAP & OVERVIEW

57.4M → 152.8M

Global Dementia Cases: 2019 to 2050

5.1M
cases in 2021

20M+
Projected by 2050

India specific

Primary Dataset: CAUEEG

- 19-channel resting-state EEG
- Clinically validated labels
- Classes: Normal, MCI, Dementia
- Includes eye open/closed annotations

Recap

- 7 papers reviewed across MRI, EEG, Audio modalities
- Selected CAUEEG dataset (Kim et al., 2023) as primary dataset
- Proposed QSFE-Net architecture (4-stream feature fusion)
- Key gap: No prior EEG work does cross-dataset validation

Research Objectives

To explore and validate the authenticity of existing EEG dementia datasets

To thoroughly analyze the CAUEEG and UCI EEG Alzheimer's datasets

To use authentic research data to uncover hidden EEG patterns linked to dementia

To increase number of clinically relevant features for dementia detection

To develop an optimized deep learning model surpassing existing performance benchmarks

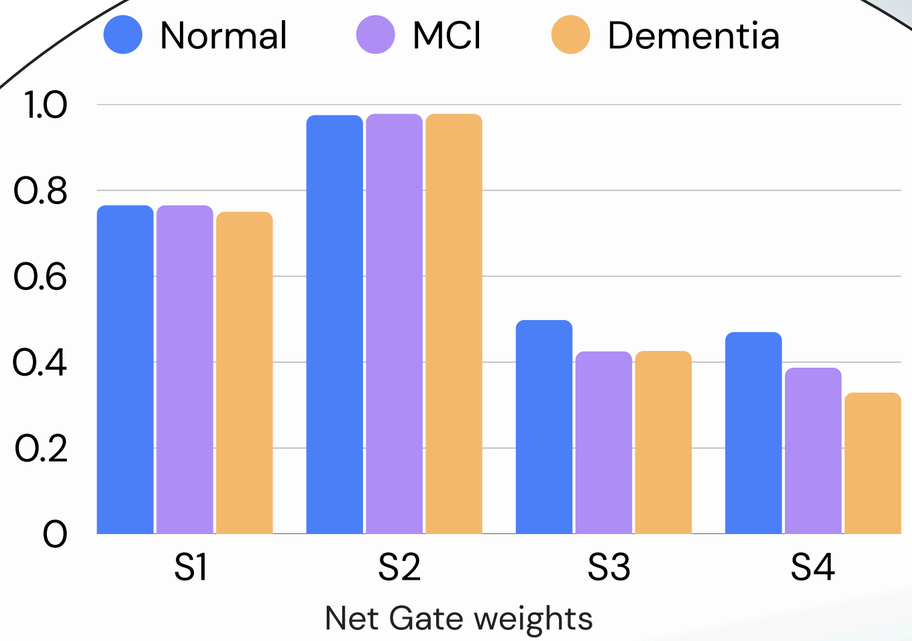
To implement existing algorithms to validate the proposed model

Research Gaps

Existing dementia models rely heavily on costly MRI systems and large architectures, while lightweight EEG-based solutions remain underexplored.

Current EEG approaches fail to effectively capture both spatial and temporal brain patterns needed for accurate dementia detection.

Most existing works lack cross-dataset validation, limiting real-world reliability and generalization.



FEATURE EXTRACTION: THE 4 STREAMS

Stream	Feature	Shape	Clinical Meaning	Separation
S1	Theta/Alpha ratio + band powers	95	Frequency slowing	Strong
S2	Multi-band coherence	684	Neural connectivity	Weak
S3	Spectral entropy	19	Signal complexity	Negligible
S4	Hemispheric asymmetry	32	Left-right imbalance	Negligible

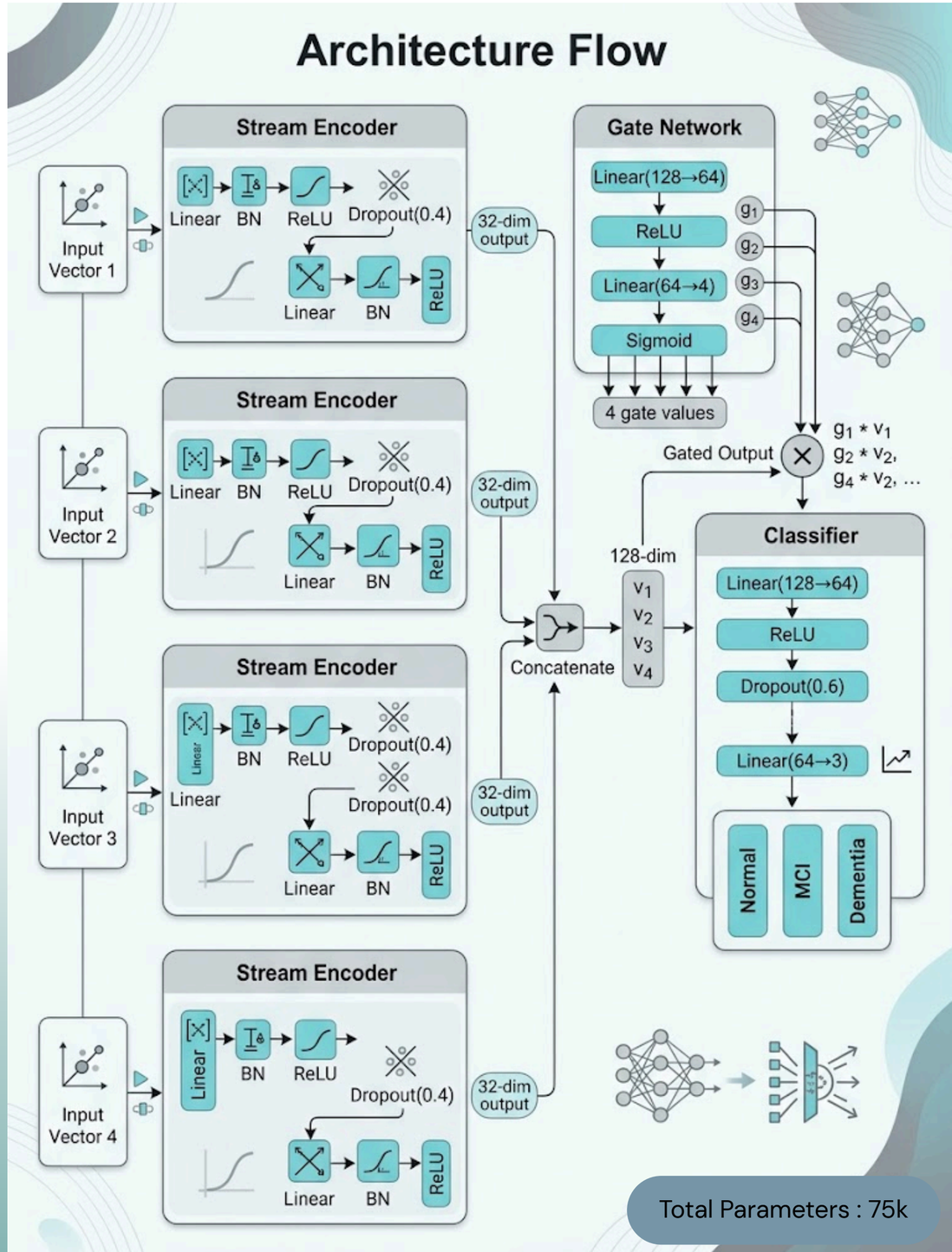
Total feature vector: 830 dimensions. Each has a clinical literature basis (cite Czigler 2008 for S1, Dauwels 2010 for S2).

Group 18:- Dementia Detection using EEG

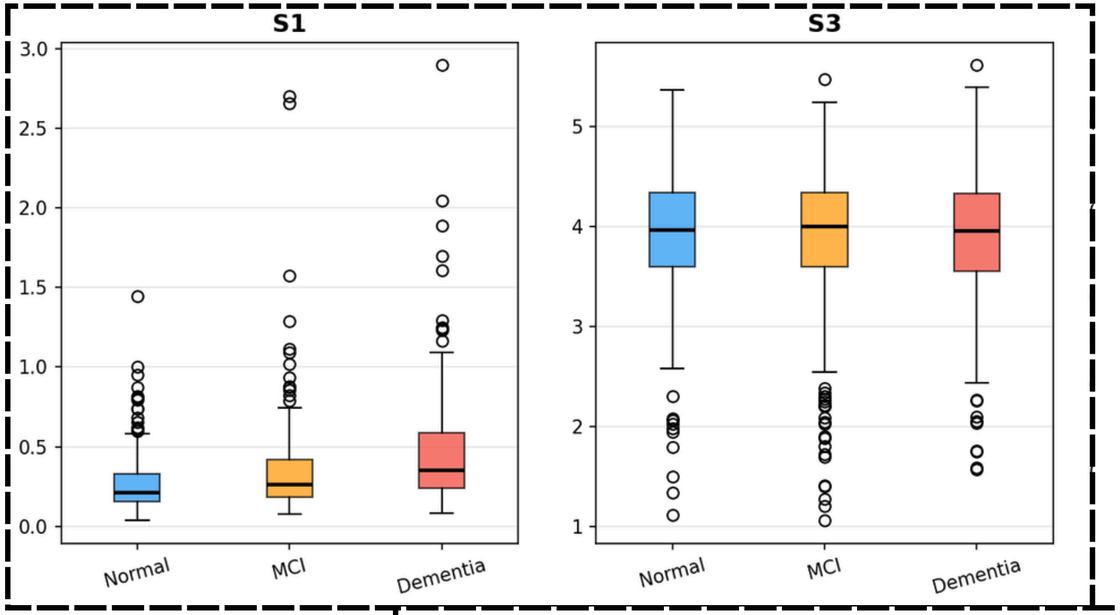
QSFE-NET ARCHITECTURE

QSFE-Net Layer Configuration

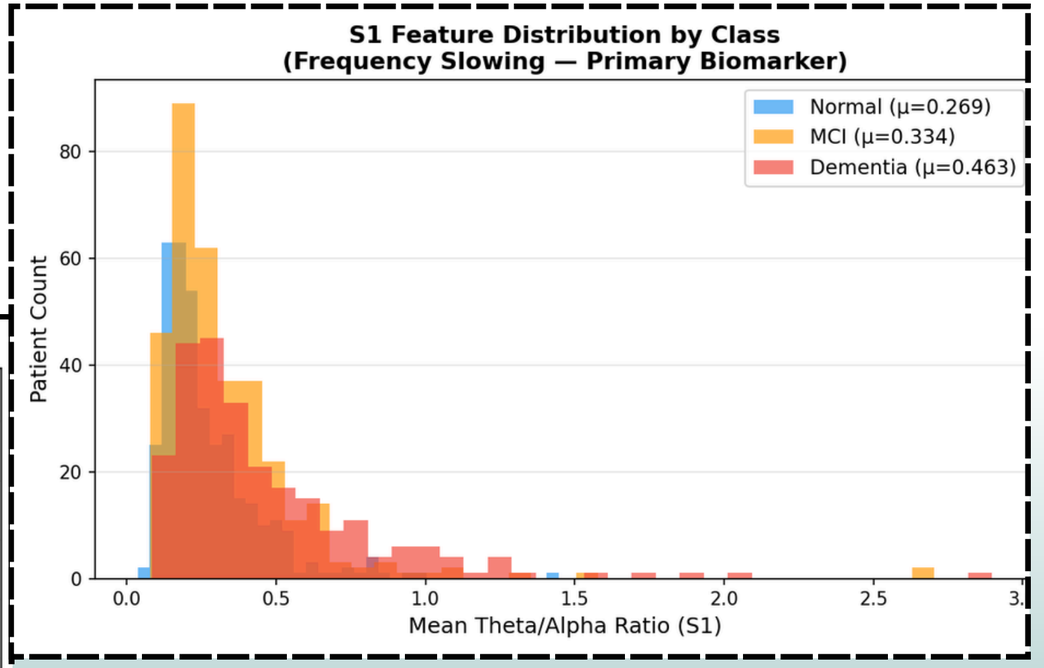
Module	Layer Configuration	Output Shape	Parameters
StreamEncoder (S1)	Linear(95 → 64) + BN + ReLU + Dropout	B × 64	~ 6,272
StreamEncoder (S1)	Linear(64 → 32) + BN + ReLU	B × 32	~ 2,112
StreamEncoder (S2)	Linear(684 → 64) + BN + ReLU + Dropout	B × 64	~ 43,968
StreamEncoder (S2)	Linear(64 → 32) + BN + ReLU	B × 32	~ 2,112
StreamEncoder (S3)	Linear(19 → 64) + BN + ReLU + Dropout	B × 64	~ 1,280
StreamEncoder (S3)	Linear(64 → 32) + BN + ReLU	B × 32	~ 2,112
StreamEncoder (S4)	Linear(32 → 64) + BN + ReLU + Dropout	B × 64	~ 2,112
StreamEncoder (S4)	Linear(64 → 32) + BN + ReLU	B × 32	~ 2,112
Gate Network	Linear(128 → 64) + ReLU + Linear(64 → 4) + Sigmoid	B × 4	~ 8,324
Classifier	Linear(128 → 64) + ReLU + Dropout + Linear(64 → 3)	B × 3	~ 8,387
Total			**79,431**

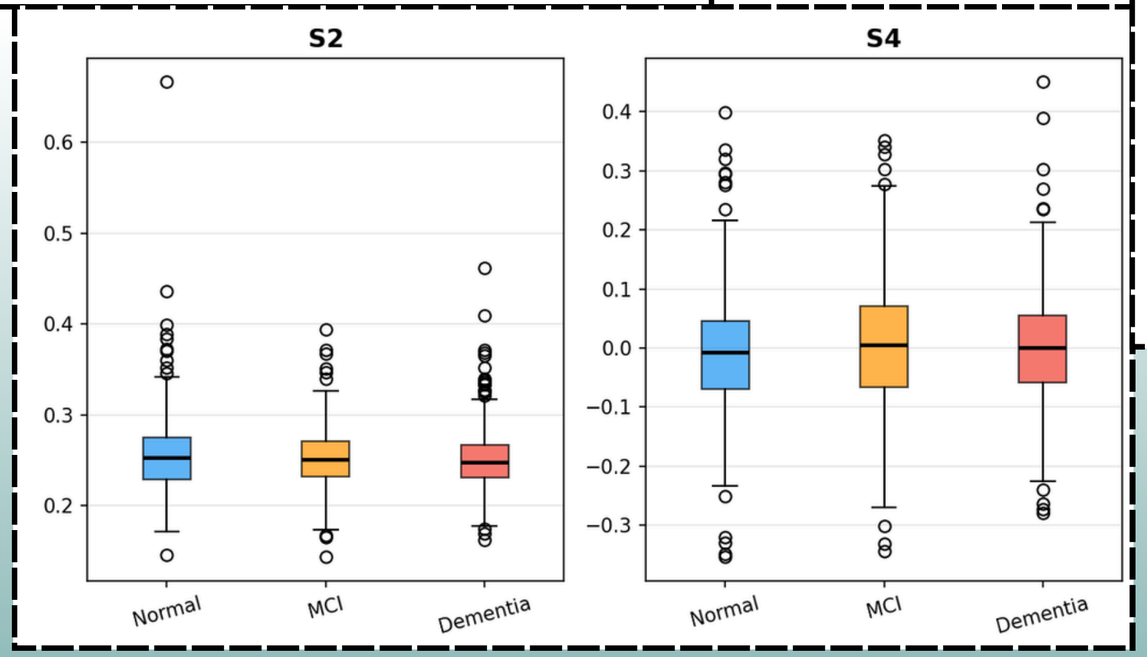


EXPLORATORY DATA ANALYSIS



S1 (strong)
⌄
S2 (weak)
⌄
S3 ≈ S4
(negligible)





Only S1 shows meaningful median shift. S2/S3/S4 distributions overlap almost entirely

S1 shows a clean 72% increase in theta/alpha ratio from Normal to Dementia — the only stream with unambiguous class separation

Group 18:- Dementia Detection using EEG

TRAINING CONFIGURATION & RESULTS

58%
Test accuracy (Run 6)

0.5626
Macro F1

1.24 pts
Val-test gap

79,431
Parameters

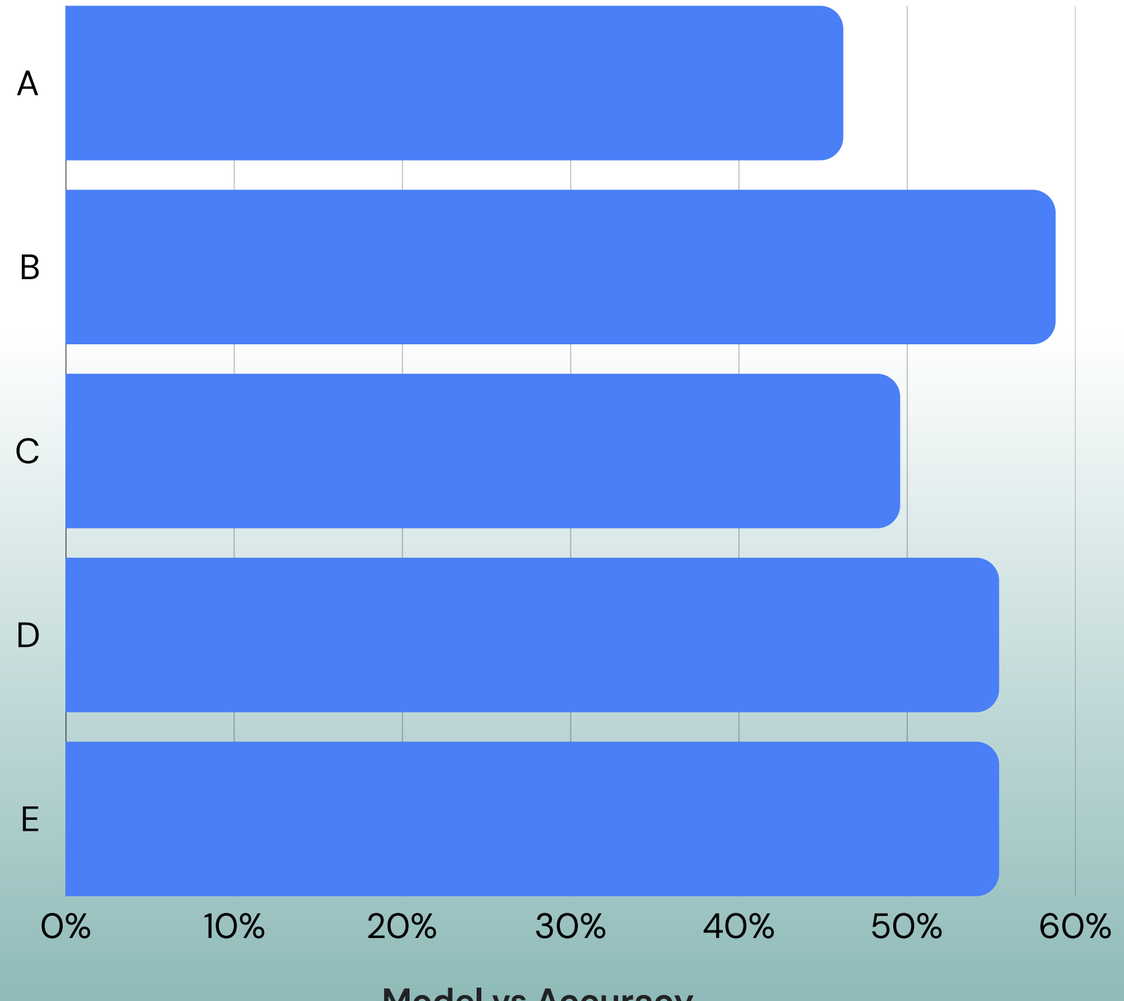
Run	Key change	Val acc	Use?
1–2	Baseline, bad features	47–54%	No
3	Fixed S2/S3/S4 features	57.98%	No (8.83pt gap)
4–6	Regularization tuning	52–54%	No
7	8 crops + best LR	58.82%	No (arch mismatch)
8	**S1 expanded (95 feat)**	**58.81%**	**Final**

CONFUSION MATRIX	Pred: Normal	Pred: MCI	Pred: Dementia
True: Normal (46)	30	13	3
True: MCI (41)	12	20	9
True: Dementia (31)	6	12	13

QSFE-Net achieved strong generalization with only a 1.24-point validation–test gap while maintaining an extremely lightweight architecture of just 79K parameters, demonstrating stable and interpretable EEG-based dementia classification.

S1+S2 is optimal. Gating provides interpretability, not accuracy gain.

ABLATION STUDY



Key Findings

Finding 1: S1 alone (46.2%) is insufficient, coherence adds significant signal (+12.6pp)

Finding 2: S3 hurts accuracy (49.58% < 58.82%) entropy introduces noise at this dataset scale (950 patients)

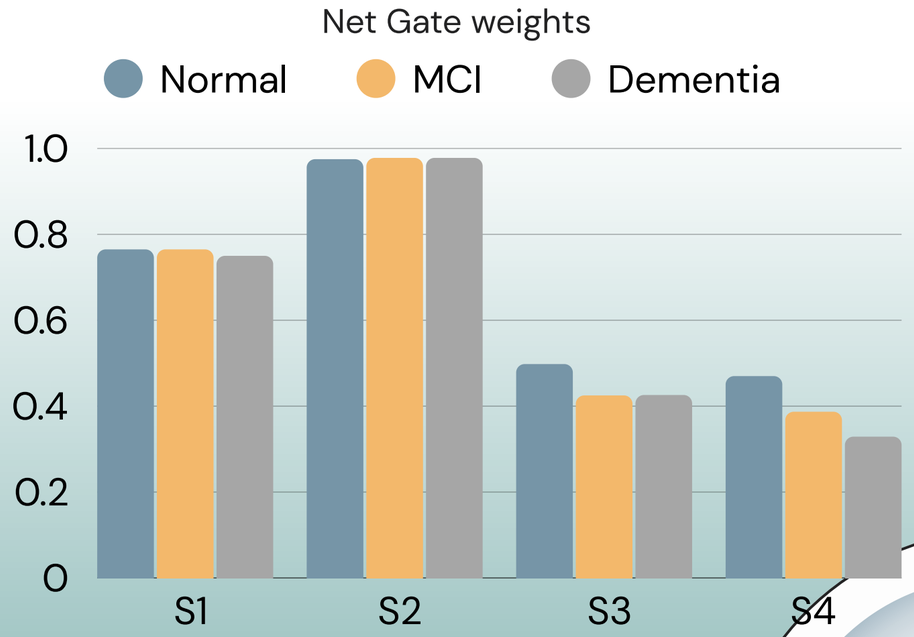
Finding 3: D = E (55.46% both), gating does not improve accuracy, but enables clinical interpretability (the gate weight story)

GATE WEIGHT ANALYSIS

Key Clinical Insight

- **S1 and S2 remain dominant biomarkers**
- **S4 shows progressive suppression from Normal → Dementia**
- **Model learned clinically meaningful patterns without explicit supervision**

* S1 (0.765→0.750): Consistently active. Primary biomarker. Slight suppression in Dementia — model shifts reliance toward coherence.
* S2 (~0.977): Fully saturated always. Coherence is universally informative even with weak global separation.
* S3 (0.498→0.425): Near-neutral for Normal, suppressed in disease. Entropy helps for healthy EEG, becomes noisy in diseased. Clinically meaningful.
* S4 (0.470→0.329): Cleanest gradient. Progressive suppression as disease worsens. Network learned asymmetry is less discriminating in severe dementia.

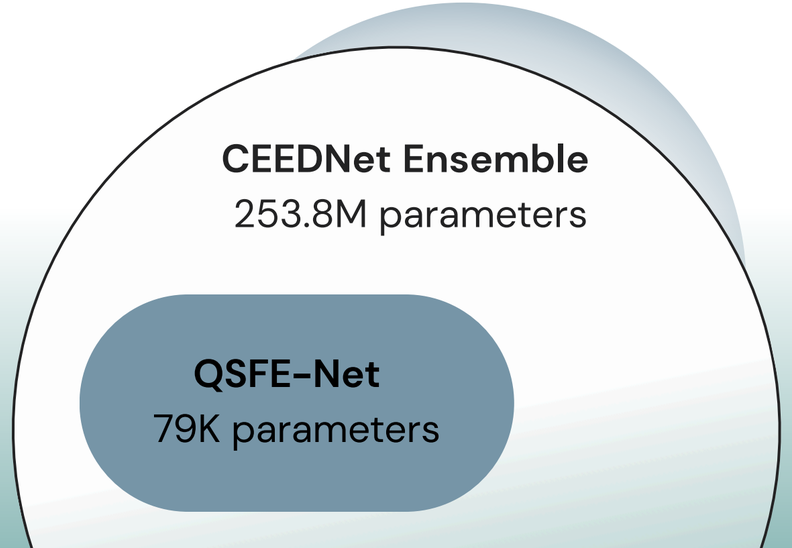


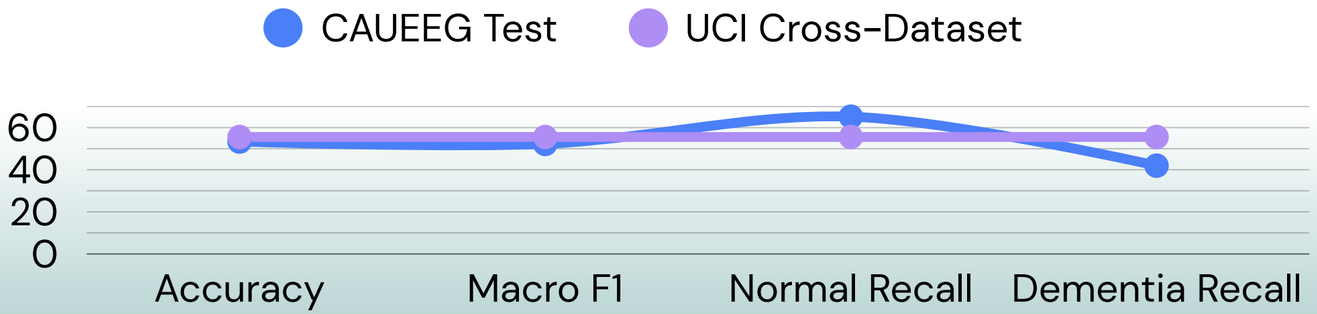
The model learned neurological disease progression. Without supervision

COMPARISON WITH EXISTING METHODS

Model	Parameters	Test Accuracy	Validation Accuracy	Macro F1	Interpretable
CEEDNet Single	25.7M	77.32%	N/A	N/A	✗
CEEDNet Ensemble	253.8M	79.16%	N/A	N/A	✗
QSFE-Net (Ours)	79,431	58.81%	58.01%	0.5626	✓

- Deployable on edge devices, portable EEG hardware, primary care settings
- Gate weights provide stream-level explanation at every prediction CEEDNet cannot





LIMITATIONS & FUTURE WORK

Known Limitations

- No artifact rejection — some recordings contain eye-blink artifacts (up to 300 μV on Fp1)
- 950 training patients is small for 3-class neurological classification — high inter-subject EEG variability
- Gating improves interpretability but not raw accuracy (D = E in ablation)
- Class imbalance: Normal > MCI > Dementia — Dementia recall is lowest (41.9%)
- Single-dataset evaluation — cross-dataset generalization untested

Immediate Next Steps:

DONE

- Architecture · 8 training runs · Ablation · EDA · Gate visualization
- Cross Validation with UCI

In Progress:

- UCI EEG Alzheimer's dataset — cross-dataset zero-shot validation
- CEEDNet baseline reimplementation on identical patient-wise splits

Planned:

- IEEE paper (gate weight analysis + ablation + cross-dataset findings)
- Frontend page for easy user access
- Artifact rejection (ICA or threshold-based)
- SMOTE oversampling
- Edge deployment on portable EEG device

THANK YOU

QSFE-Net

A Lightweight & Interpretable EEG-Based Dementia Classification Framework

REFERENCES

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Key Project Highlights

- *79,431 parameter lightweight architecture*
- *3-class EEG classification (Normal / MCI / Dementia)*
- *Interpretable gate-weight based predictions*
- *3,200× smaller than CEEDNet Ensemble*